Evaluating the major causes of Carbon Emissions in Chittagong Port: An overview

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Abstract

Chittagong port is considered the biggest port in Bangladesh. That is why the city is also known as 'The Port City. But this port is responsible for major air pollution and carbon emission inside the city. Yearly carbon emission from Chittagong Port is about 29.5 lac tons whereas the annual carbon emission rate in Port of New York and New Jersey is about 113001 tons and port of Long Beach is about 115792 tons. compared to other ports outside the country, Chittagong port in Bangladesh has a huge range of carbon emissions that can be a threat to the city dwellers and the country. The data is based on secondary data analysis aimed at finding the major causes and factors of carbon emission from Chittagong port. Carbon dioxide, Carbon monoxide, and black carbon are taken into consideration. Causes mainly focused on transportation: Head trucks, Lorry, Locomotives, and Cargo handling equipment. According to World Development Indicators (World Bank Database), Carbon Emission in Bangladesh for 2019 is estimated at 0.6 metric tons per capita which was 0.4 metric tons per capita in 2014. That means the carbon emission rate in Bangladesh is increasing day by day. This study also recommends some suggestions on lowering carbon emissions to have a pollution-free and sustainable port and also a sustainable city like Chattogram.

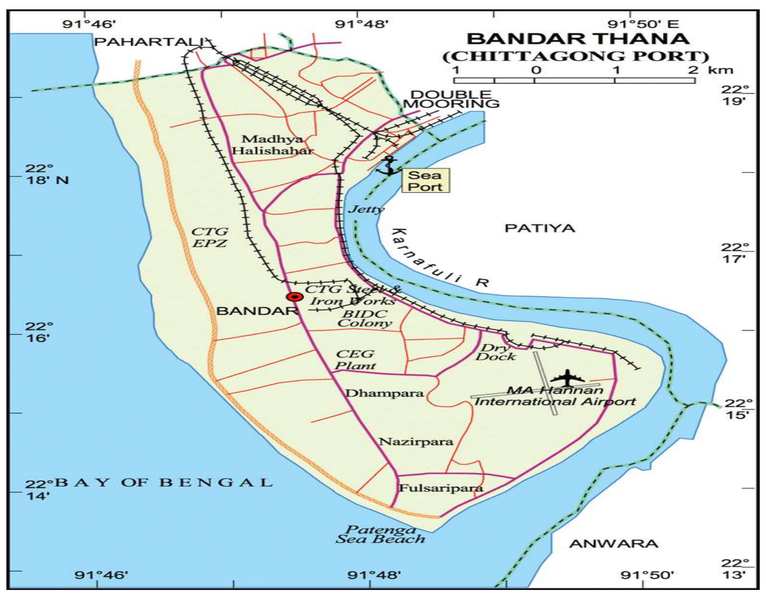
Introduction

Chittagong port is holding a significant part of Chittagong city for contributing to the economy of the country. Chittagong port is used by India, Nepal, and Bhutan for transshipment. About 92 percent of export-import of Bangladesh largely depended on Chittagong Port. But the port also contributing to the major carbon emission inside the city.The main culprit for the major carbon emission in Chittagong port is the transports that includes Head trucks, Locomotives, marine vessels, Cargo handling equipment, and container vehicles as they use diesel, oil, LNG, and methanol as fuel that produces more carbon. This study reveals the major carbon emission of Chittagong port focusing on Carbon dioxides, Carbon Mono-oxides and Black carbon. It is responsible for temperature expansion and sea level rise in the city. Also, that causes major health impacts such as respiratory diseases and inhalation problems in the residents. So it's time to take proper initiatives to control carbon emissions along the area to have a sustainable port that will make an important role in the economy of the country, considering the environmental factors

Methodology

- The study is based on secondary data analysis. A good number of Journals, Books, Articles, recent news reports, and publications are analyzed for having an admissible conclusion.The carbo emission rate in Bangladesh has been taken from the World Development Indicator (World bank database).
- The Chittagong port is located on the bank of Karnaphuli, which is almost 15 kilometers from the open Bay of Bengal. The geographical locational value of Chittagong port is LAT 22 13.54 N and Long 91 48 (Panocean, 2019). According to Lloyd's Chittagong port is ranked as the world's 58th busiest port (Barua, 2020).

Source: ctgreflection.blogspot.com



Results

Carbon Emimision Status:

Chittagong port has a significant value for carbon emission in Chittagong city as the port uses a big number of trucks, container vessels, locomotives, and Container handling equipment that emits Carbon dioxide, Carbon monoxide, and black carbon.

Head Trucks: Heavy duty vehicles like trucks produce more carbon as they use carbon-containing fuel diesel and burn fuels for long times. Chittagong port has two types of head trucks: Light Container Vehicle and Heavy Container Vehicle. Head trucks of Chittagong port emit 5075580.60 tons of CO_2 1075.61 tons of Carbon Mono-oxides (CO), and 14.44 tons of black carbon. Total emissions from the light Container Vehicle are 3758034.50 TONNES OF CO_2, 795.44 tons of CO, and 10.72 tons of black carbon. Total emission from Heavy Container vehicles is 1317546.10 tons of CO_2, 280.17 tons of CO, and 3.73 tons of black carbon (Akther et al., 2020)

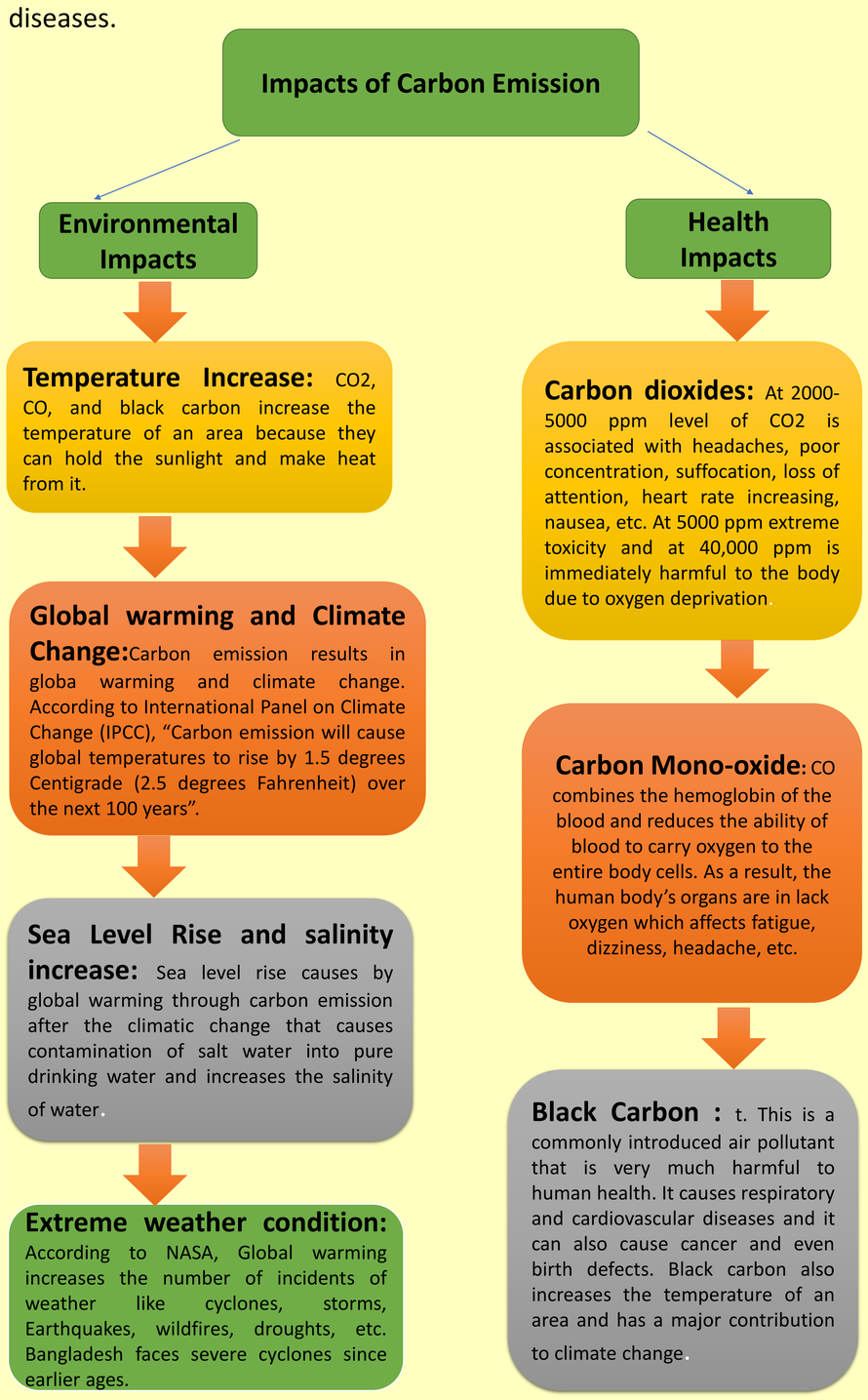
Container Handling Equipment: For handling of container in Chittagong port many different types of equipment are used: Container mover, Forklift, Forklift truck, Heavy trailer, Industrial Tractor, Log Handler, Mobile Crane, Mobile harbor crane, Pick and Carry, Quay Gantry Crane, Reach Stacker and Rubber Tyred Grantyl crane, Straddle Carrier and Terminal Tractor, etc. The total emission of CO_2 is 64633.59 tons, CO is 57.15 tons and black carbon is 9.17 tons(Akther et al., 2020)

Locomotives: Locomotives used in Chittagong port are very old, built in 1953. Locomotives have both port and port emissions. The total per month emission of CO_2 is 63.19 tons, CO is 0.12 and black carbon is 63.19 tons. Total emission per year from Locomotives: CO_2 is 758.30, CO is 1.47 and black carbon is 0.27. Emission from Locomotives per trip: CO_2 is 9.03 tons, CO is 0.02 tons and black carbon is 0.003 tons(Akther et al., 2020).

Environmental and Health Impacts:

Carbon emissions especially Carbon dioxide emissions play a vital role to increase the surface temperature of the earth. Carbon dioxide acts like a blanket for trapping solar temperature from the sun to the surface. According to U.S Environmental Protection Agency, Carbon emission especially Carbon dioxide emission causes 80% of greenhouse gas emissions in the US (Cairoli, 2018). That causes environmental impacts such as water supplies, weather patterns, crop growing season changes, and increased sea level rise in the coastal areas.

Besides, The increasing amount of emission of carbon that is predominantly focused on Carbon dioxides and other Carbon containing pollutants can harm human health. According to World Health Organization*(WHO), 'Climate change is projected to increase the percentage of people in Mali suffering from hunger from 34% to 64%, about 40 years from now. Moreover, Carbon containing pollutants are directly associated with respiratory and vector-borne diseases.



Recommendations

Reducing carbon footprint: First of all, the Chittagong Port Authority (CPA) needs to be alert about its carbon footprint and make alert all levels of the worker. They should be alerted about their energy use. Every worker should be educated about carbon footprint and the uses of energy and fuels used in trucks and equipment. The authority should use renewable energy sources for their activities of container handling and official works. The defective transport and equipment should be replaced for less carbon emission.

Tree plantation inside the port and the city: The Chittagong Port Authority (CPA) should take an initiative to plant more trees inside the port. They should make places for at least shrub plants inside the port and the government should plant more and more trees on the highways and roadsides so that the trucks carrying goods outside the port to other districts, carbon emissions can be reduced as trees and plants can absorb Carbon, especially Carbon dioxide.

Alternative Fuel usage: Generally, the road transports of Chittagong port including trucks, Locomotives, and containers handling vehicles and equipment use diesel as fuel which is liquid fuel. In Bangladesh, liquid fuels are mainly used for transportation(Islam et al., 2012). According to Bengtsson, Fridell & Andersson,2012, The ships which are used to carry containers usually use Liquified Natural Gas (LNG) and Methanol which can be used potentially as a low carbon fuel for ships and transports at the port (Winnes et al., 2015). For low carbon emissions, the CPA authority should use dual fuel engines for road transports, ships, and equipment that can use either fuel oil or LNG or other low carbon emission-based gas.

Policy Implications

Chittagong Port Authority (CPA) and the Government of Bangladesh should adopt some plans and policies and implement them properly to reduce carbon emissions inside and outside the port. Some recommended policies are as below:

1. To take a mitigation policy/plan strategically and apply it systematically to reduce carbon emissions from the Chittagong port area.
2. Change the fuel usage from diesel to less carbon-emitting fuels for trucks, ships, and types of equipment and use renewable energy sources, making an on-shore power supply to reduce carbon emission.
3. To conduct some complementary projects on specifically Chittagong port's environmental issues and promote research and practical study to have a sustainable port.
4. Practicing clean and green revolution inside and outside the surrounding area of the port.
5. To remove old and useless equipment and transport, make a budget for new and eco-friendly equipment and transport for the port
6. To collect appropriate and accurate data about air quality and carbon emission of the port for a specific period and data should be updated over time.
7. Government initiatives must be followed to reduce Carbon emissions and escape from global warming and climate change difficulties . (Sarkar, 2018

Conclusion Remarks

Chittagong port is considered a gateway to Bangladesh's foreign trade and economy. We are using the port since 1887. For our country's economic betterment, we need the port to be sustainable for hundreds and thousands of years more. That is why we need to assure less carbon emission from the port, especially from the transportation sector. The government of Bangladesh (GOB) and the Chittagong Port Authority (CPA) need to develop a mitigation policy and implement their strategies to make the port a carbon emission-free sustainable port for the betterment of the people living and working there, and also to protect the city from excessive temperature increase, sea level rise and the bad impacts of global warming and climate change.

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PP-24 Tobacco Cultivation, Farmer Dependance Health and Environment

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The tobacco cultivation has been starting massively from the early 1990's and various national and international tobacco companies have been investing there, and now most of the tobacco cultivators producing tobacco are associated with the company. According to Bangladesh Bureu of Statistics, the union "Refaitpur" located in Dulatpur Upazila is one of the highest tobacco cultivation places in Bangladesh. This is a mix method study, where in-depth interviews, focus group discussions, KII were used for primary data collection. The thematic data analysis process has used for data analysis along with "Microsoft office and Atlis.ti" in many cases. Total 35 in-depth interviews and 5 five FGD and 7 kII have conducted for the study. Most of the respondent expose that tobacco farming is comparatively economically beneficial than other crops grown in the study area, but in the long run its creating a vulnerable situation. All most 73% respondent agreed that, in the long run due to tobacco cultivation. Tobacco cultivation is a combined process which enhanced by the socio-economic condition, environmental benefit, earning raw money as well as other external factors are also highly responsible for that. Prior to tobacco cultivation extensively it was a totally grain cultivated area, but after this area known throughout the country for tobacco cultivation. Although in the initial stage it could be seen that tobacco cultivating is highly profitable for the farm owner but in the long run it makes the farmer more marginalized. This study tried to find out how the tobacco companies are making marginal farmers through the contractual farming process in addition, its impact on health and environment also.

PP-25 Evaluating the Major Causes of Carbon Emissions in Chittagong Port: An Overview

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